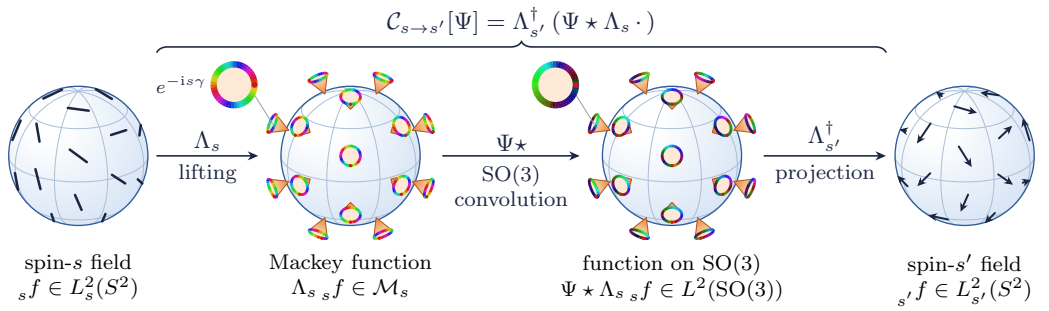
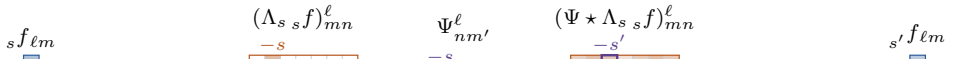

$\mathcal{C}_{s\to s'}[\Psi] = \Lambda^\dagger_{s'}\,(\Psi \star \Lambda_s\,\cdot\,)$
$e^{-is\gamma}$
Λ_s
lifting
$\Psi\star$
SO(3)
convolution
$\Lambda^\dagger_{s'}$
projection
spin-s field
${}_sf \in L^2_s(S^2)$
Mackey function
$\Lambda_s\,{}_sf \in \mathcal{M}_s$
function on SO(3)
$\Psi \star \Lambda_s\,{}_sf \in L^2(\mathrm{SO}(3))$
spin-s' field
${}_{s'}f \in L^2_{s'}(S^2)$



$$\left(\mathcal{C}_{s \to s'}[\Psi] \; {}_s f\right)_{\ell m} = (-1)^{s+s'} \, \overline{\Psi^{\ell}_{-s',-s}} \; {}_s f_{\ell m}$$

	output on S^2 (project, $\Lambda_{s'}^{\dagger}$)	output on SO(3) (no projection)
input on S^2 (lift, Λ_s)	 $\mathcal{C}_{s\to s'}$ entry $(-s', -s)$ axisymmetric, left convolution, spin-weighted, ð, E/B, Kaiser–Squires cost $\mathcal{O}(L^3)$	 $\mathcal{C}_{s\to G}$ column $m' = -s$ directional (scalar and spin), steerable when $\lvert n\rvert < N$ cost $\mathcal{O}(NL^3)$
input on SO(3) (no lift)	 $\mathcal{C}_{G\to s'}$ row $n = -s'$ synthesis (transpose) convolution, fibre average when $s' = 0$ cost $\mathcal{O}(NL^3)$	 $\mathcal{C}_{G\to G}$ full block group convolution on SO(3) cost $\mathcal{O}(L^4)$